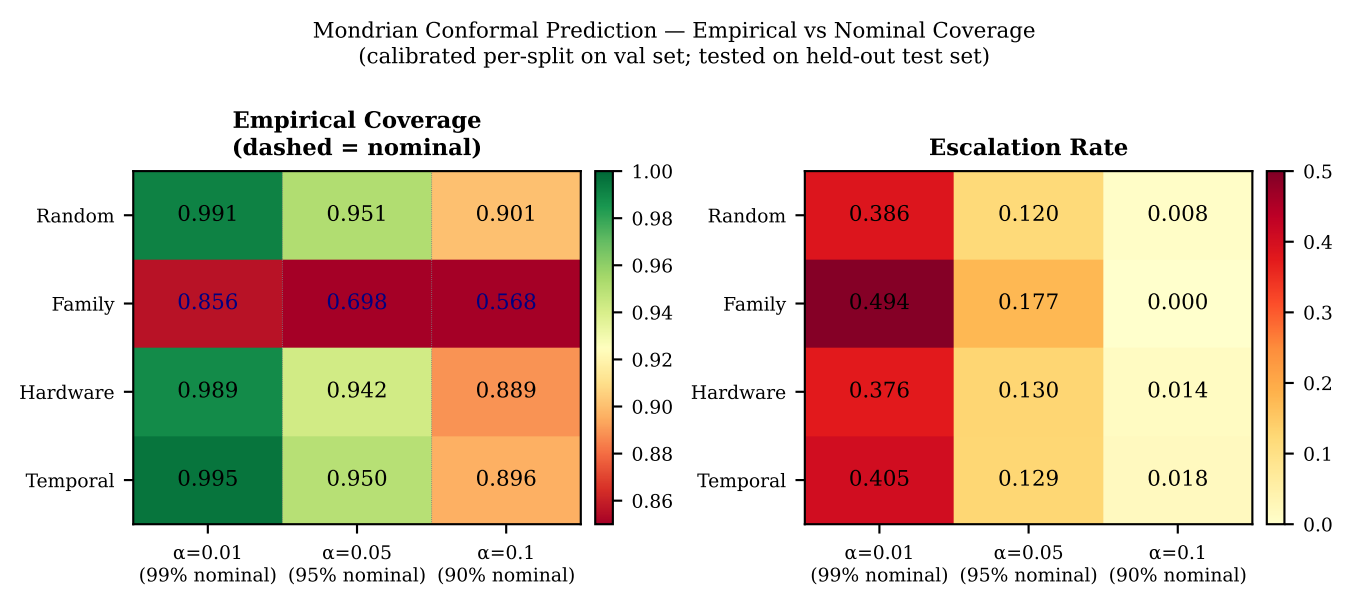
Mondrian Conformal Prediction — Empirical vs Nominal Coverage
(calibrated per-split on val set; tested on held-out test set)
Empirical Coverage
(dashed = nominal)
Random
Family
Hardware
Temporal
0.991
0.951
0.901
0.856
0.698
0.568
0.989
0.942
0.889
0.995
0.950
0.896
1.00
0.98
0.96
0.94
0.92
0.90
0.88
0.86
α=0.01
(99% nominal)
α=0.05
(95% nominal)
α=0.1
(90% nominal)
Escalation Rate
Random
Family
Hardware
Temporal
0.386
0.120
0.008
0.494
0.177
0.000
0.376
0.130
0.014
0.405
0.129
0.018
0.5
0.4
0.3
0.2
0.1
0.0
α=0.01
(99% nominal)
α=0.05
(95% nominal)
α=0.1
(90% nominal)